HW4

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Problem 1

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'tidyr' was built under R version 4.4.3

Warning: package 'readr' was built under R version 4.4.3

Warning: package 'purrr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.3

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.0     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.0
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(nzelect)
```

a.Generate a table (which can just be a nicely printed tibble) of vote count (regardless of party) per year/type. Make sure to sort it by vote count.

```
nzge %>%
  group_by(voting_type, election_year) %>%
  summarize(votes = sum(votes)) %>%
  ungroup() %>%
  arrange(desc(votes))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by voting_type and election_year.
i Output is grouped by voting_type.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(voting_type, election_year))` for per-operation
  grouping (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 10 x 3
   voting_type election_year   votes
   <chr>               <dbl>   <dbl>
 1 Party                2014 2416479
 2 Candidate            2014 2375493
 3 Party                2008 2356536
 4 Candidate            2008 2325598
 5 Party                2005 2286190
 6 Candidate            2005 2260670
 7 Party                2011 2257336
 8 Candidate            2011 2225766
 9 Party                2002 2040248
10 Candidate            2002 2022115
```

b. Focus only on the 2014 election. Report the proportion of votes for each party in the Candidate election. Again, produce a nice table and sort it by percent of vote.

```
nzge %>%
```

```r
  # keep only candidate votes from the 2014 election
  filter(election_year == 2014 & voting_type == "Candidate") %>%

  # total number of votes for each party
  group_by(party) %>%
  summarize(votes = sum(votes)) %>%
  ungroup() %>%

  arrange(desc(votes)) %>%

  # each party's proportion of the total Candidate vote
  mutate(total = sum(votes),
         prop = round(votes/total*100, 3)) %>%

  select(-total, -votes)
```

```
# A tibble: 25 x 2
   party                      prop
   <chr>                     <dbl>
 1 National Party            45.5
 2 Labour Party              33.7
 3 Green Party                6.98
 4 Conservative Party         3.41
 5 New Zealand First Party    3.09
 6 Maori Party                1.77
 7 MANA Movement              1.36
 8 Informal Candidate Votes   1.17
 9 ACT New Zealand            1.17
10 United Future              0.62
# i 15 more rows
```

c.Produce a nice table indicating, for each year, which party won the Candidate vote and which party won the Party vote.

```r
nzge %>%
  group_by(voting_type, election_year, party) %>%
  summarize(votes = sum(votes)) %>%

  # find the party with the highest vote count in each election
  arrange(election_year, voting_type, desc(votes)) %>%
```

```
  slice(1) %>%
  select(-votes) %>%

  # reshape the results Candidate and Party appear in separate columns
  pivot_wider(names_from = voting_type,
              values_from = party)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by voting_type, election_year, and party.
i Output is grouped by voting_type and election_year.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(voting_type, election_year, party))` for per-operation
  grouping (`?dplyr::dplyr_by`) instead.

# A tibble: 5 x 3
# Groups:   election_year [5]
  election_year Candidate      Party
          <dbl> <chr>          <chr>
1          2002 Labour Party   Labour Party
2          2005 National Party Labour Party
3          2008 National Party National Party
4          2011 National Party National Party
5          2014 National Party National Party
```

Problem 2-Tidyverse: Tennis

```
library(tidyverse)

tennis <- read_csv(
  "https://raw.githubusercontent.com/farhadGithub/tennis-atp-data/main/data/raw/atp_matches_
)
```

```
Rows: 2806 Columns: 49
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (14): tourney_id, tourney_name, surface, tourney_level, winner_entry, wi...
dbl (35): draw_size, tourney_date, match_num, winner_id, winner_seed, winner...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

a. How many tournaments took place in 2019?

```
tournaments <- tennis %>%
  group_by(tourney_name) %>%
  summarize() %>%
  ungroup()
tournaments %>%
  nrow()
```

```
[1] 128
```

```
# check the data
tournaments %>%
  filter(str_detect(tourney_name, "Davis Cup")) %>%
  slice(1:5)
```

```
# A tibble: 5 x 1
  tourney_name
  <chr>
1 Davis Cup Finals F: CAN vs ESP
2 Davis Cup Finals QF: ARG vs ESP
3 Davis Cup Finals QF: AUS vs CAN
4 Davis Cup Finals QF: GBR vs GER
5 Davis Cup Finals QF: SRB vs RUS
```

```
tournaments <- tournaments %>%
  mutate(tourney_name = str_replace(tourney_name, "Davis.*", "Davis Cup")) %>%
  group_by(tourney_name) %>%
  summarize() %>%
  ungroup()
tournaments %>%
  nrow()
```

```
[1] 69
```

There are only 69 unique tournaments.

b.Did any player win more than one tournament? If so, how many players won more than one tournament, and how many tournaments did the most winning player(s) win?

```
winners <- tennis %>%
  arrange(tourney_name, desc(match_num)) %>%
  group_by(tourney_name) %>%
  slice(1) %>%
  ungroup() %>%
  select(winner_name) %>%
  group_by(winner_name) %>%
  summarize(wins = n()) %>%
  ungroup() %>%
  arrange(desc(wins)) %>%
  # identify players who won more than one tournament
  filter(wins > 1)
winners %>%
  nrow()
```

```
[1] 17
```

```
winners %>%
  slice(1:2)
```

```
# A tibble: 2 x 2
  winner_name     wins
  <chr>          <int>
1 Rafael Nadal       9
2 Novak Djokovic     8
```

There are 17 players who won multiple tournaments. And Rafael Nadal won the most, 9 tournaments.

c. Is there any evidence that winners have more aces than losers? (If you address this with a hypothesis test, do not use base R functionality - continue to remain in the Tidyverse.)

```
library(tidymodels)
```

```
-- Attaching packages ------------------------------------- tidymodels 1.4.1 --

v broom        1.0.12     v rsample      1.3.2
v dials        1.4.2      v tailor       0.1.0
v infer        1.1.0      v tune         2.0.1
v modeldata    1.5.1      v workflows    1.3.0
v parsnip      1.4.1      v workflowsets 1.1.1
v recipes      1.3.1      v yardstick    1.3.2

Warning: package 'broom' was built under R version 4.4.3

Warning: package 'infer' was built under R version 4.4.3

Warning: package 'parsnip' was built under R version 4.4.3

Warning: package 'rsample' was built under R version 4.4.3

-- Conflicts ---------------------------------------- tidymodels_conflicts() --
x scales::discard() masks purrr::discard()
x dplyr::filter()   masks stats::filter()
x recipes::fixed()  masks stringr::fixed()
x dplyr::lag()      masks stats::lag()
x yardstick::spec() masks readr::spec()
x recipes::step()   masks stats::step()
```

```r
tennis %>%
  select(w_ace, l_ace) %>%
  mutate(win_more = w_ace > l_ace) %>%
  prop_test(win_more ~ NULL, p = 0.5)
```

```
# A tibble: 1 x 4
  statistic chisq_df  p_value alternative
      <dbl>    <int>    <dbl> <chr>
1      56.7        1 4.95e-14 two.sided
```

Yes, we have statistical significant evidence that winners perform more aces than losers.

d. Identify the player(s) with the highest win-rate. (Note that this is NOT asking for the highest number of wins.) Restrict to players with at least 5 matches.

```r
tennis %>%
  select(winner_name, loser_name) %>%
  pivot_longer(everything(),
               names_to = "outcome",
               values_to = "player") %>%

  # whether each player won the match
  mutate(winner = outcome == "winner_name") %>%
  # number of matches and win rate for each player
  group_by(player) %>%
  summarize(matches = n(),
            winrate = mean(winner)) %>%
  ungroup() %>%
  # players who played at least 5 matches
  filter(matches >= 5) %>%

  # sort players from highest to lowest win rate
  arrange(desc(winrate)) %>%
  slice(1:2)
```

```
# A tibble: 2 x 3
  player         matches winrate
  <chr>            <int>   <dbl>
1 Rafael Nadal        69   0.870
2 Novak Djokovic      69   0.841
```

Only one player has the highest win-rate, and it is Rafael Nadal.

Problem 3 - Visualization

```r
covid <- read.csv("https://raw.githubusercontent.com/nytimes/covid-19-data/refs/heads/master/
str(covid)
```

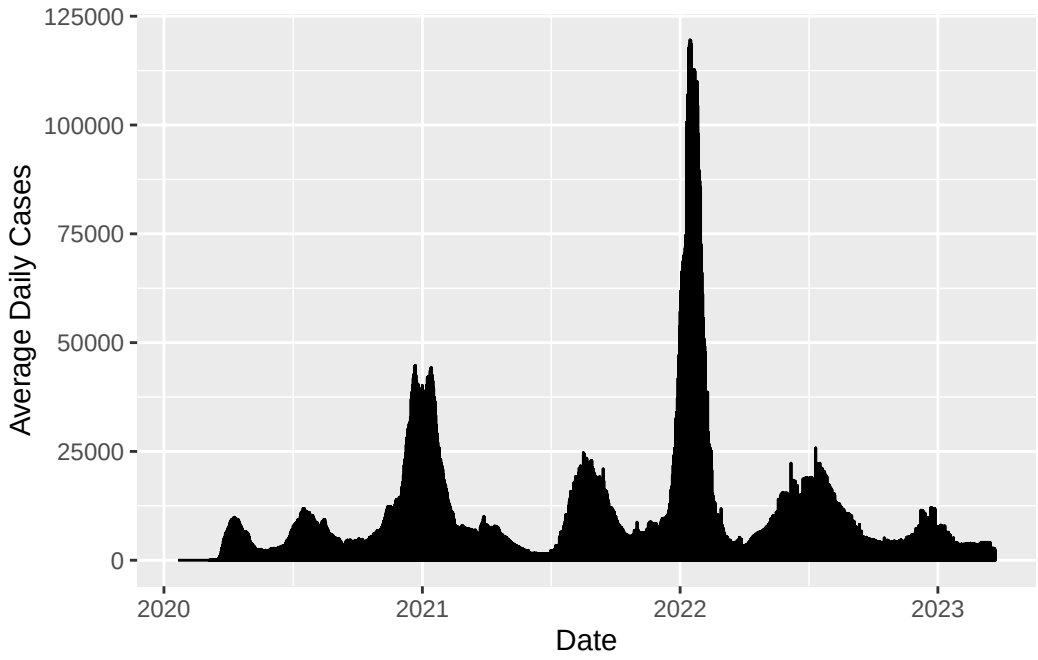
```
'data.frame':   61942 obs. of  9 variables:
 $ date              : chr  "2020-01-21" "2020-01-22" "2020-01-23" "2020-01-
24" ...
 $ geoid             : chr  "USA-53" "USA-53" "USA-53" "USA-53" ...
 $ state             : chr  "Washington" "Washington" "Washington" "Washington" ...
 $ cases             : int  1 0 0 0 1 0 0 1 0 0 ...
```

```
 $ cases_avg          : num  0.14 0.14 0.14 0.14 0.14 0.14 0.14 0.14 0.14 0.14 ...
 $ cases_avg_per_100k : num  0 0 0 0 0 0 0 0 0 0 ...
 $ deaths             : int  0 0 0 0 0 0 0 0 0 0 ...
 $ deaths_avg         : num  0 0 0 0 0 0 0 0 0 0 ...
 $ deaths_avg_per_100k: num  0 0 0 0 0 0 0 0 0 0 ...
```

```
covid$date <- lubridate::ymd(covid$date)
names(covid)[names(covid) == "date"] <- "Date"
names(covid)[names(covid) == "state"] <- "State"
```

a. How many major and minor spikes in cases were there?

```
library(ggplot2)
ggplot(covid, aes(x = Date, y = cases_avg)) +
  geom_line() +
  ylab("Average Daily Cases")
```

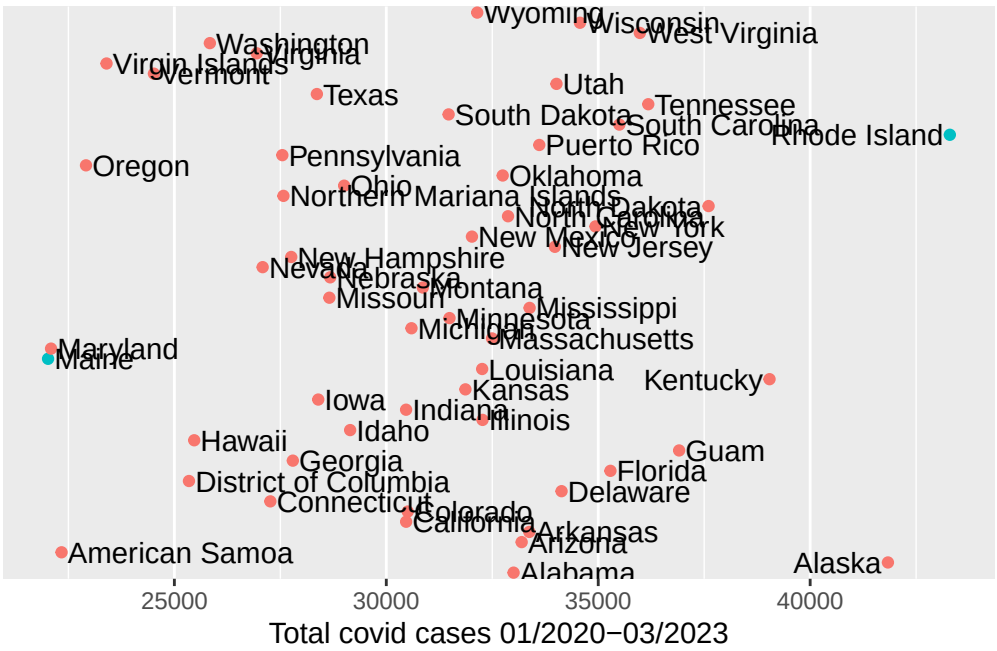


The plot shows approximately four major spikes and three or four smaller spikes.

b. For the states with the highest and lowest overall rates per population, what differences do you see in their trajectories over time?

```
# aggregate to the state level
covid_state <- covid %>%
  group_by(State) %>%
  summarize(total = sum(cases_avg_per_100k)) %>%
  ungroup()
```

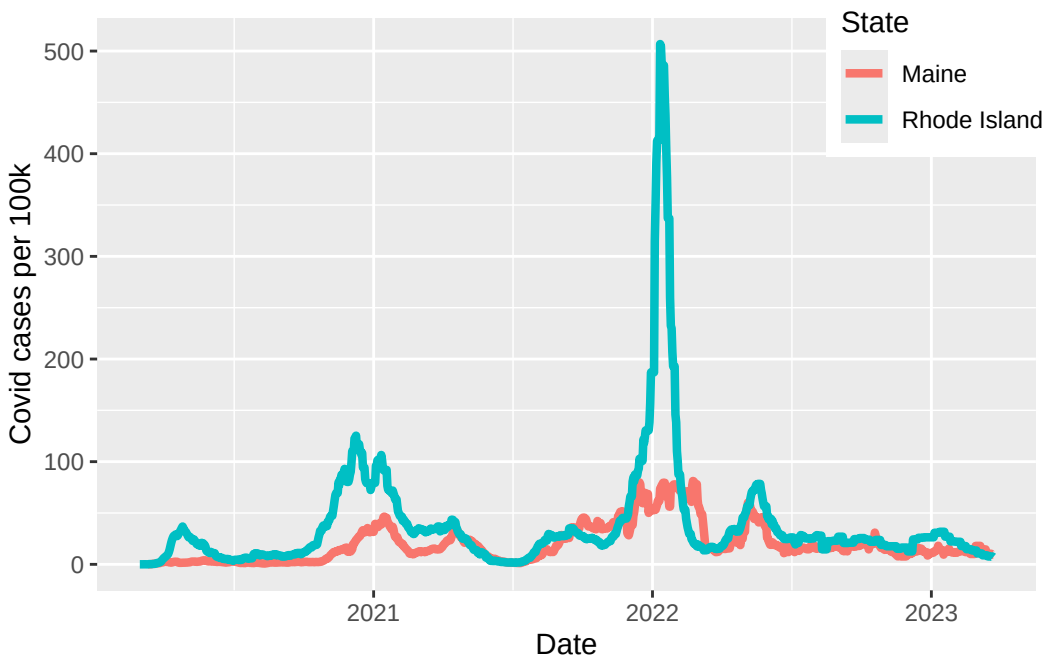
```
ggplot(covid_state, aes(x = total, y = State, label = State)) +
  geom_point(aes(color = total == max(total) | total == min(total))) +
  geom_text(aes(nudge_x = ifelse(total > 37500, -150, 150),
                hjust = ifelse(total > 37500, 1, 0))) +
  scale_y_discrete(labels = NULL, breaks = NULL, name = "") +
  theme(legend.position = "off") +
  xlab("Total covid cases 01/2020-03/2023")
```



Rhode Island has the highest overall case rate, while Maine has the lowest.

```
ggplot(covid %>% filter(State == "Maine" | State == "Rhode Island"),
       aes(x = Date, y = cases_avg_per_100k, color = State)) +
  geom_line(linewidth = 1.5) +
```

```
  ylab("Covid cases per 100k") +
  theme(legend.position = "inside", legend.position.inside = c(.9, .9))
```



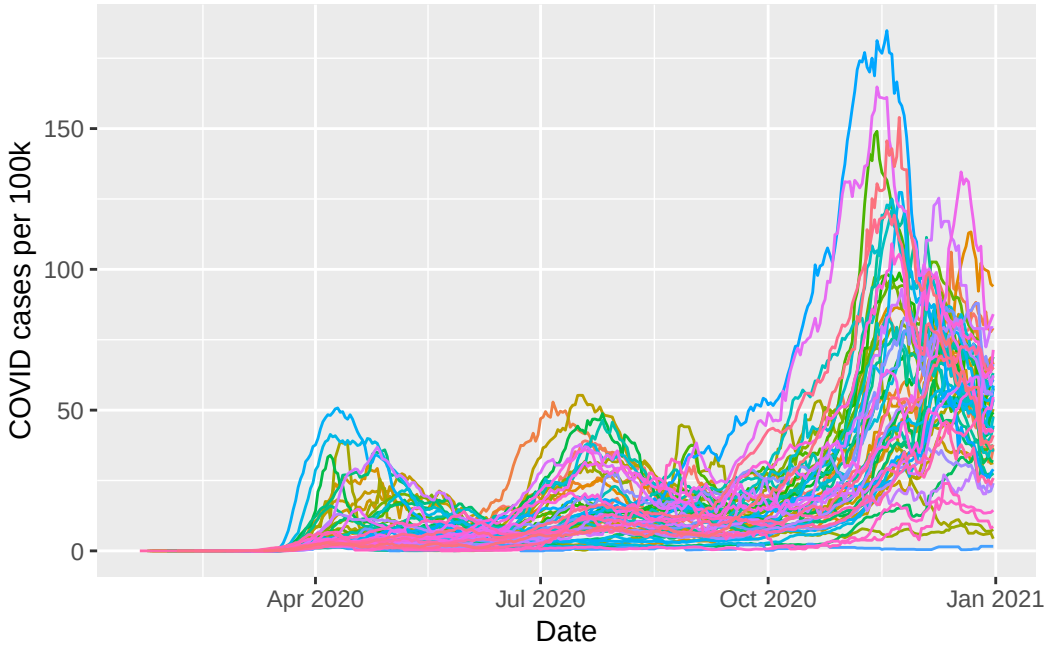
Rhode Island generally had higher case rates than Maine and experienced much sharper spikes, particularly around early 2022. Maine's trajectory was comparatively smoother, with smaller peaks throughout most of the pandemic. And the spike in winter 21-22 did not seem to affect Maine, while increasing rates around 10-fold in Rhode Island.

ChatGPT helped me figure out a more briefly way to show my result.

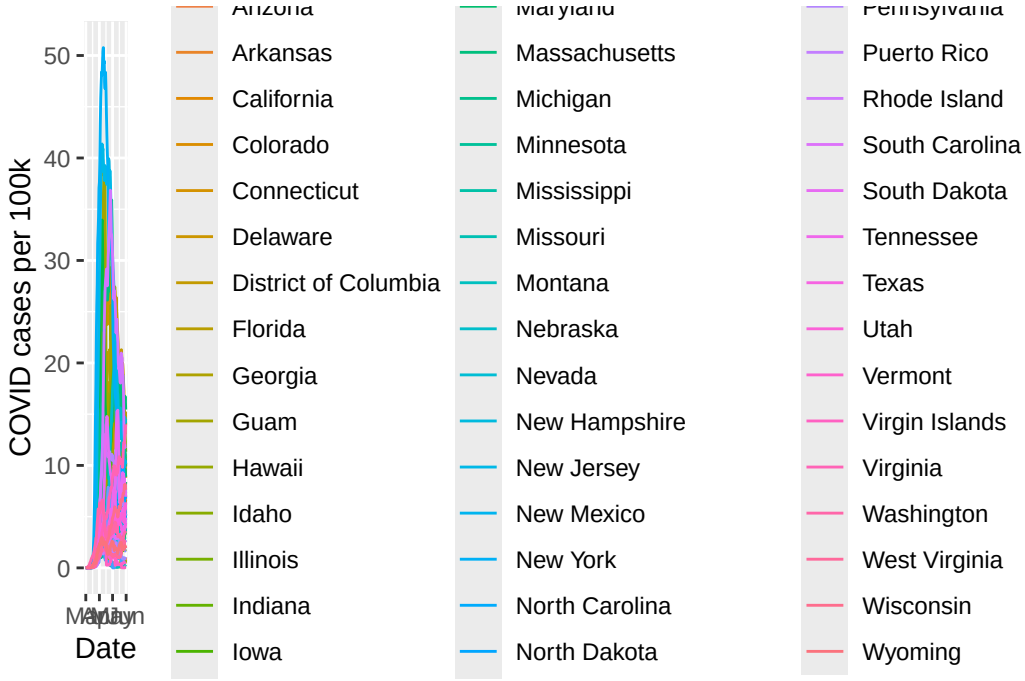
c. Identify, to the best of your ability without a formal test, the first five states to experience Covid in a substantial way.

```
covid20 <- covid %>%
  filter(year(Date) == 2020) %>%
  select(State, Date, cases_avg_per_100k)
```

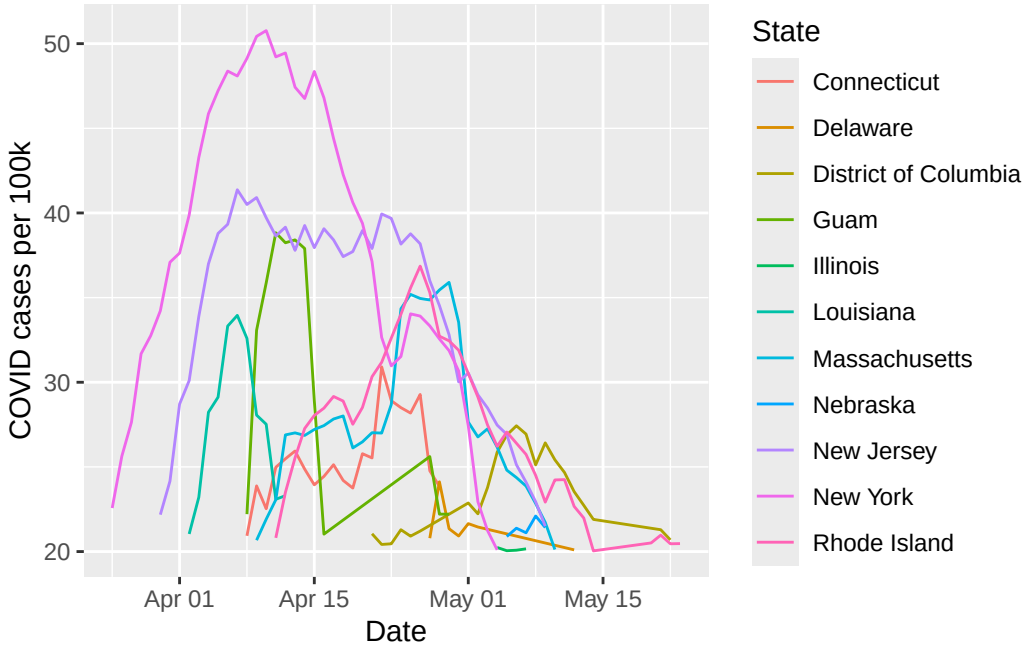
```
ggplot(covid20, aes(x = Date, y = cases_avg_per_100k, color = State)) +
  geom_line() +
  ylab("COVID cases per 100k") +
  theme(legend.position = "off")
```




```
covid20 <- covid20 %>%
  filter(month(Date) %in% c(3, 4, 5)) # Mar-May
ggplot(covid20, aes(x = Date, y = cases_avg_per_100k, color = State)) +
  geom_line() +
  ylab("COVID cases per 100k")
```




```
# that legend is overwhelming, restrict data where the cases above 20
ggplot(covid20 %>% filter(cases_avg_per_100k > 20),
       aes(x = Date, y = cases_avg_per_100k, color = State)) +
  ylab("COVID cases per 100k") +
  geom_line()
```



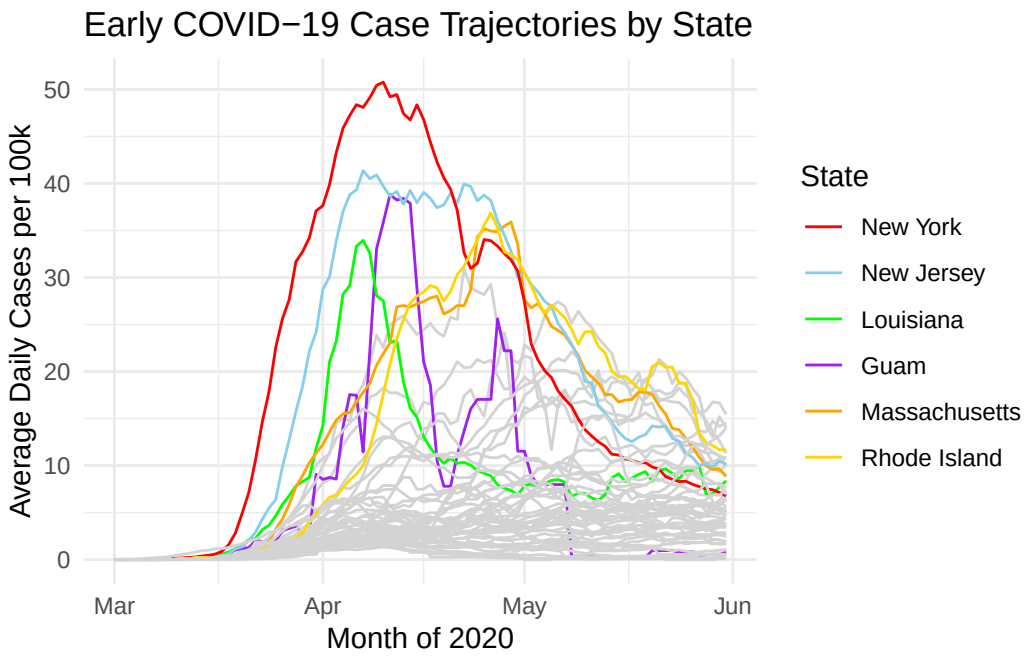

```
# states that experienced substantial COVID rates earliest
topstates <- c(
  "New York",
  "New Jersey",
  "Louisiana",
  "Guam",
  "Massachusetts",
  "Rhode Island"
)

# assign gray to all states and distinct colors to the selected states
all_states <- unique(covid20$State)

colors <- rep("lightgrey", length(all_states))
names(colors) <- all_states

colors[topstates[1]] <- "red"
colors[topstates[2]] <- "skyblue"
colors[topstates[3]] <- "green"
colors[topstates[4]] <- "purple"
colors[topstates[5]] <- "orange"
colors[topstates[6]] <- "gold"
```

```r
# highlight the states with the earliest substantial increases
ggplot(
  covid20,
  aes(x = Date, y = cases_avg_per_100k, color = State)
) +
  geom_line() +
  scale_color_manual(
    values = colors,
    breaks = topstates
  ) +
  labs(
    title = "Early COVID-19 Case Trajectories by State",
    x = "Month of 2020",
    y = "Average Daily Cases per 100k",
    color = "State"
  ) +
  theme_minimal()
```



Based on the early trajectories from March to May 2020, New York, New Jersey, Louisiana, and Guam clearly experienced substantial COVID-19 case rates relatively early. Massachusetts and Rhode Island also showed early increases and are difficult to distinguish as the fifth state, so both are highlighted in the final plot.